

Stochastic Modeling and Analysis of Segregated Fund Guarantees: GMWB, GLWB and GMIB

Stanislas MBOUNGOU MANGOUBI

Abstract

This technical document presents an exhaustive modeling of segregated fund guarantees (GMWB, GLWB, GMIB) within a risk-neutral framework $\mathbb{Q}$. We detail the valuation equations from the perspective of the insurer (reserves) and the policyholder (fair value), as well as the underlying financial models used to project interest rates and risky assets.

1. CONTEXT AND DATA

The IFRS 17 Challenge: The IFRS 17 standard requires a *Market-Consistent* valuation of insurance liabilities. For segregated funds, this involves capturing the Time Value of Options and Guarantees (TVOG) via stochastic simulations.

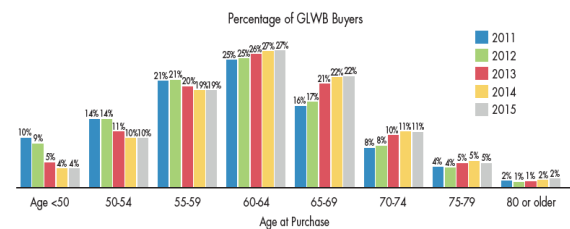
Data (SOA 2016): The study is based on the analysis of 4.9 million contracts.

- **GLWB/GMIB:** 90% of the in-force guarantee market.
- **Inertia:** Once withdrawals begin, they tend to continue (27% utilization for GLWB).
- **Lapse (Surrender):** Strong correlation with utilization (3.4% global vs 10% if withdrawals are active).

Modeling Objectives:

1. Implement a risk-neutral Economic Scenario Generator (ESG).
2. Develop payoff functions for each product.
3. Calculate the average present value of future cash flows (Best Estimate Provisions).

Figure 1-1: GLWB Buyers by Age at Time of Purchase, 2011–2015



Note: Based on 1,524,669 contracts issued between 2011 and 2015, and still in force at EOY 2015.

Figure 1: Concentration of GLWB buyers (Source: SOA).

2. UNDERLYING FINANCIAL MODELS ($\mathbb{Q}$)

We use a hybrid framework to project the economy, calibrated to market data.

Interest Rate Model (Vasicek)

The dynamics of the instantaneous short rate r_t are governed by a stochastic differential equation (SDE) with mean reversion:

$$dr_t = a(b - r_t)dt + \sigma_r dW_t^r \quad (1)$$

Where a is the speed of mean reversion, b is the long-term mean level, and σ_r is the volatility. The price of a zero-coupon bond $P(t, T)$ is given by the affine form:

$$P(t, T) = A(t, T)e^{-B(t, T)r_t} \quad (2)$$

$$B(t, T) = \frac{1 - e^{-a(T-t)}}{a} \quad (3)$$

Fund Dynamics (Black-Scholes with Fees)

The value of the segregated fund A_t evolves according to the performance of the index S_t minus continuous management fees φ (MER):

$$\frac{dS_t}{S_t} = r_t dt + \sigma_S dW_t^S \quad (4)$$

Correlation assumption: We consider independence between interest rates and stocks. This implies:

$$Cov(W_t^S, W_t^r) = 0 \quad (5)$$

Discrete evolution of the personal account:

$$A_{t_{i+1}} = A_{t_i} \times e^{-\varphi \Delta t} \times \frac{S_{t_{i+1}}}{S_{t_i}} \quad (6)$$

3. GMWB (Guaranteed Minimum Withdrawal Benefit)

The GMWB guarantees the repayment of the initial capital via periodic withdrawals over a fixed duration M , regardless of the fund's performance.

3.1 Cash Flow Mechanics

Withdrawal Flow and Fund Evolution

Let T_r be the retirement age. The guaranteed withdrawal W_{t_k} is defined by:

$$W_{t_k} = \begin{cases} 0, & t_k < T_r \\ G_{T_r} \times x_W, & T_r \leq t_k < T_r + M \end{cases} \quad (7)$$

At each withdrawal date, the fund is reduced by the withdrawal amount (down to 0):

$$A_{t_k}^+ = \max(A_{t_k}^- - W_{t_k}, 0) \quad (8)$$

Step-up Mechanism (Cliquet): The guarantee base G_t can increase if the market performs well during the accumulation phase:

$$G_{t_k} = \max(G_{t_{k-1}}, A_{t_k}), \quad G_0 = A_0 \quad (9)$$

3.2 Liability Valuation (Insurer)

The insurer must cover the difference if the guaranteed withdrawal exceeds the value of the available fund.

Insurer's Cost and Liability

The cash outflow for the insurer at time t_k is:

$$C_{t_k} = \begin{cases} 0, & t_k < T_r \\ \max(W_{t_k} - A_{t_k}, 0), & T_r \leq t_k < T_r + M \end{cases} \quad (10)$$

The value of the liability $\Pi_t^{Insurer}$ is the difference between expected costs and collected fees:

$$\begin{aligned} \Pi_t^{Insurer} = & \underbrace{\sum_{i=1}^M (T_r - t + i) p_{x+t} \cdot \mathbb{E}^Q \left[e^{-\int_t^{T_r+i-1} r_u du} C_{(T_r+i-1)} \right]}_{\text{Guarantee Costs (Outflows)}} \\ & - \underbrace{\sum_{j=1}^{M+T_r-t} j p_{x+t} \cdot \mathbb{E}^Q \left[e^{-\int_t^{t+j} r_u du} (1 - e^{-\varphi}) A_{t+j} \right]}_{\text{Management Fees (Inflows)}} \end{aligned} \quad (11)$$

3.3 Fair Value for the Policyholder

The policyholder values all potential benefits (withdrawals, residual value, death benefit).

Total Value for the Policyholder

$$\begin{aligned}
 \Pi_t^{\text{Policyholder}} = & \underbrace{\sum_{i=1}^M (T_r - t + i - 1) p_{x+t} \times \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^{T_r-i-1} r_u du} W_{(T_r+i-1)} | F_t \right]}_{\text{1. Guaranteed Withdrawals}} \\
 & + \underbrace{(M + T_r - t - 1) p_{x+t} \times \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^{M+T_r-1} r_u du} A_{M+T_r-1} | F_t \right]}_{\text{2. Residual Fund Value (if survival)}} \\
 & + \underbrace{\sum_{i=1}^{M+T_r-1} (i-1) p_{x+t} \times q_{x+t+(i-1)}^{(d)} \times \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^{t+i} r_u du} A_{t+i} | F_t \right]}_{\text{3. Death Benefit}} \\
 & + \underbrace{\sum_{i=1}^{M+T_r-1} (i-1) p_{x+t} \times q_{x+t+(i-1)}^{(w)} \times \mathbb{E}^{\mathbb{Q}} \left[e^{-\int_t^{t+i} r_u du} A_{t+i} | F_t \right]}_{\text{4. Surrender Value (Lapse)}}
 \end{aligned} \tag{12}$$

4. GLWB (Guaranteed Lifetime Withdrawal Benefit)

The GLWB is an extension of the GMWB where withdrawals are guaranteed **for life**. This introduces a major longevity risk for the insurer.

4.1 Lifetime Flows

Lifetime Guarantee

Unlike the GMWB, the withdrawal period is not capped by M but extends until death (limiting age ω):

$$W_{t_k} = \begin{cases} 0, & t_k < T_r \\ G_{T_r} \times x_W, & t_k \geq T_r \quad (\forall t_k \text{ up to } \omega) \end{cases} \tag{13}$$

The cost for the insurer arises when the fund is depleted ($A_t = 0$), transforming the product into a pure life annuity funded by the insurer.

4.2 Liability Valuation (Insurer)

The summation now extends over the entire possible residual lifetime.

GLWB Liability

$$\begin{aligned}\Pi_t^{Insurer} = & \sum_{i=1}^{\omega-x-t} (T_r-t+i)p_{x+t} \cdot \mathbb{E}^Q \left[e^{-\int_t^{T_r+i-1} r_u du} \max(W_{t_i} - A_{t_i}, 0) \right] \\ & - \sum_{j=1}^{\omega-x-t} j p_{x+t} \cdot \mathbb{E}^Q \left[e^{-\int_t^{t+j} r_u du} (1 - e^{-\varphi}) A_{t+j} \right]\end{aligned}\quad (14)$$

Note: The first term represents a deferred life annuity conditional on fund depletion.

4.3 Fair Value for the Policyholder

The equation for the policyholder is similar to the GMWB but with a time horizon ω .

Total Value GLWB

$$\begin{aligned}\Pi_t^{\text{Policyholder}} = & \underbrace{\sum_{i=1}^{\omega-x-t} i-1 p_{x+t} \cdot \mathbb{E}^Q \left[e^{-\int_t^{t+i-1} r_u du} W_{t+i-1} \mathbb{I}_{\{t+i-1 \geq T_r\}} \right]}_{\text{1. Guaranteed Withdrawals (Life Annuity)}} \\ & + \underbrace{\omega-x-t p_{x+t} \cdot \mathbb{E}^Q \left[e^{-\int_t^{\omega-x} r_u du} A_{\omega-x} \right]}_{\text{2. Residual Value (if max survival)}} \\ & + \underbrace{\sum_{i=1}^{\omega-x-t} i-1 p_{x+t} \cdot q_{x+t+i-1}^{(d)} \cdot \mathbb{E}^Q \left[e^{-\int_t^{t+i} r_u du} A_{t+i} \right]}_{\text{3. Death Benefit (Capital to beneficiary)}} \\ & + \underbrace{\sum_{i=1}^{\omega-x-t} i-1 p_{x+t} \cdot q_{x+t+i-1}^{(w)} \cdot \mathbb{E}^Q \left[e^{-\int_t^{t+i} r_u du} A_{t+i} \right]}_{\text{4. Surrender Value (Lapse)}}\end{aligned}\quad (15)$$

5. GMIB (Guaranteed Minimum Income Benefit)

The GMIB offers an option to convert capital into a life annuity at a guaranteed rate, protecting the policyholder against a market downturn and/or a drop in interest rates at retirement.

5.1 Guarantee Base and Option**Guarantee Base (Roll-up & Ratchet)**

The guarantee base G_t^W is the greater of a fixed compound return (i_w) and the maximum historical fund value (Ratchet):

$$G_t^W = \max \left(A_0(1 + i_w)^t, \max_{0 \leq t_k \leq t} (A_{t_k}) \right) \quad (16)$$

Retirement Decision (T_r)

At retirement, the policyholder exercises the option if the actuarial value of the guaranteed annuity exceeds their account value:

$$W_{T_r} = \max(A_{T_r}, G_{T_r}^W \times g \times a(T_r)) \quad (17)$$

Where g is the guaranteed conversion rate and $a(T_r)$ is the market price of a \$1 life annuity at time T_r :

$$a(T_r) = \sum_{k=1}^{\omega-(x+T_r)} k p_{x+T_r}^{(d)} \cdot P(T_r, T_r + k) \quad (18)$$

5.2 Liability Valuation (Insurer)

The cost for the insurer is one-off (at time T_r), unlike the periodic flows of the GMWB/GLWB.

GMIB Liability

$$\begin{aligned} \Pi_t^{Insurer} = & \underbrace{T_{r-t} p_{x+t} \cdot \mathbb{E}^Q \left[e^{-\int_t^{T_r} r_u du} \max(G_{T_r}^W \cdot g \cdot a(T_r) - A_{T_r}, 0) \right]}_{\text{Cost of the Exchange Option}} \\ & - \sum_{i=1}^{T_r-t} i p_{x+t} \cdot \mathbb{E}^Q \left[e^{-\int_t^{t+i} r_u du} (1 - e^{-\varphi}) A_{t+i} \right] \end{aligned} \quad (19)$$

5.3 Fair Value for the Policyholder**Total Value GMIB**

$$\begin{aligned} \Pi_t^{\text{Policyholder}} = & T_{r-t} p_{x+t} \cdot \mathbb{E}^Q \left[e^{-\int_t^{T_r} r_u du} \max(G_{T_r}^W \cdot g \cdot a(T_r), A_{T_r}) \right] \quad (\text{Retirement Option}) \\ & + \sum_{j=1}^{T_r-t} j-1 p_{x+t} \cdot q_{x+t+j-1}^{(d)} \cdot \mathbb{E}^Q \left[e^{-\int_t^{t+j} r_u du} A_{t+j} \right] \quad (\text{Death}) \\ & + \sum_{j=1}^{T_r-t} j-1 p_{x+t} \cdot q_{x+t+j-1}^{(w)} \cdot \mathbb{E}^Q \left[e^{-\int_t^{t+j} r_u du} A_{t+j} \right] \quad (\text{Surrender}) \end{aligned} \quad (20)$$

6. MODELING ASSUMPTIONS

The valuations rely on rigorous calibration reflecting a risk-neutral economic environment and best actuarial estimates.

Table 1: Summary of Base Assumptions

Parameter	Assumption / Model
Mortality	CPM-2014 Table (Projected 2024, Scale B)
Lapse (Surrender)	3.4% global (Differentiated: 10% if active withdrawals, 3% otherwise)
Risk-Free Rate (r_0)	5.0% (Base Scenario), Vasicek dynamics
Equity Volatility (σ)	20% (Base), Black-Scholes model
Management Fees (MER)	2.0% per year (deducted continuously)
Initial Fund (A_0)	\$100 000 (Single Premium)
Age at Issue	55 years (Start of withdrawals at 65)

Technical Note: The lapse assumption is critical. In accordance with the SOA study, we apply a *partial rational lapse*: policyholders using their guarantees (active withdrawals) have a much lower propensity to surrender (3× lower) than those not using the option.

7. SYNTHESIS OF RESULTS AND SENSITIVITIES

The table below presents the average Fair Value of contracts under different stress scenarios, calculated on an initial fund of \$100,000.

Table 2: Summary Table: Simple and Composite Shocks

Type	Scenario	GMWB	GLWB	GMIB
Ref.	Base ($r = 5\%, \sigma = 20\%$)	82 271	76 530	100 965
Simple	Low Rates ($r = 1\%$)	83 289	80 382	109 604
	High Volatility ($\sigma = 30\%$)	84 495	84 721	120 429
	Strong Retention (Lapse -5%)	75 520	71 428	106 138
	Longevity Shock ($q_x - 1\%$)	81 855	76 056	101 423
Composite	Global Shock ($\sigma \uparrow$ and $r \downarrow$)	84 495	84 721	120 429
	Behavior (Retention + Longevity)	74 994	71 317	106 866

Analysis of Results:

- **GMIB and Rates:** The GMIB is extremely sensitive to low rates. In the global shock scenario, its value increases by 20% compared to the base case, as the conversion option becomes deeply "In-the-money".
- **Lapse Paradox:** For GMWB and GLWB, better retention (lower lapses) *decreases* the fair value of the contract for the policyholder (from 82k to 75k). By staying invested, the policyholder pays fees (MER) over a longer period, which benefits the insurer.

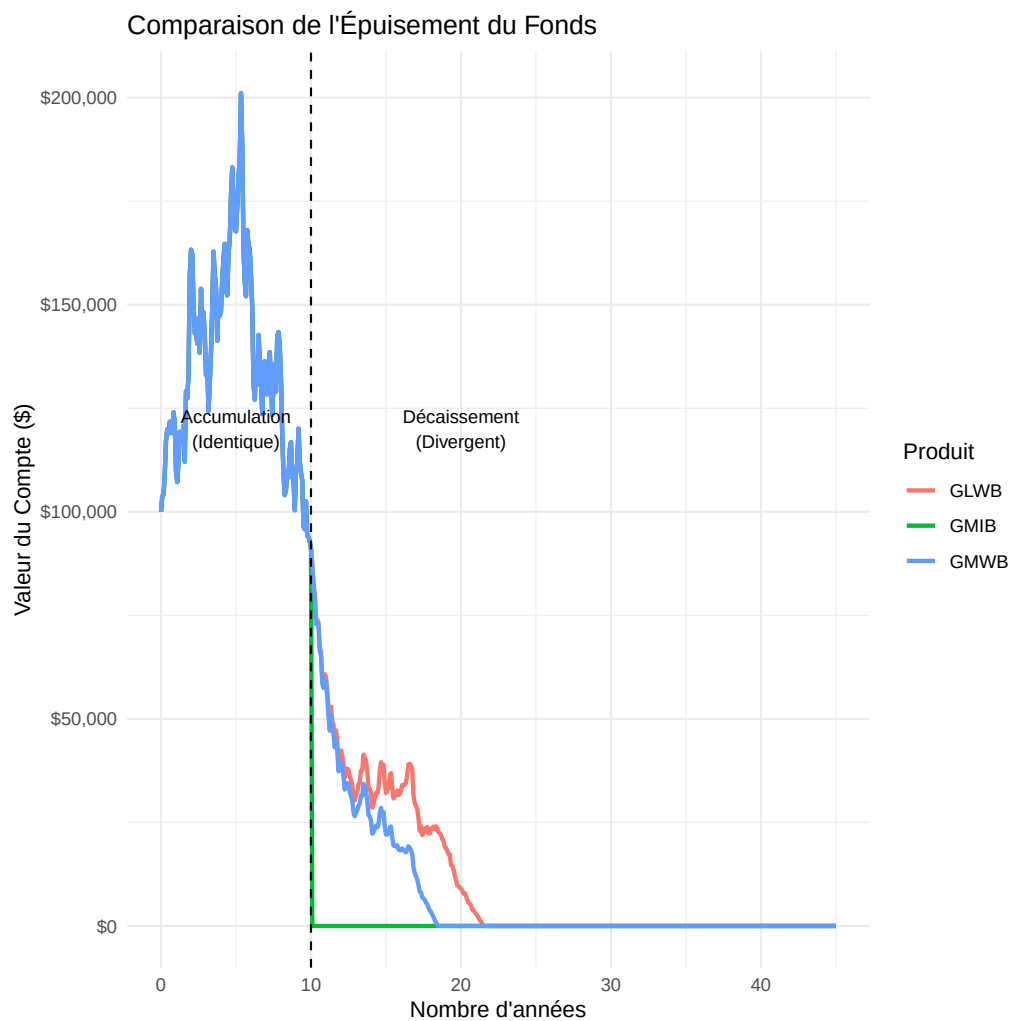


Figure 2: Decumulation dynamics: The GMIB drops to zero at retirement (conversion), while GMWB/GLWB depletes gradually.

7. CONCLUSION AND LIMITATIONS

This work unifies the modeling of segregated funds under IFRS 17.

- **Rationality:** The model assumes static rationality. A dynamic approach ("Rational Lapse") would increase the liability, as policyholders would keep the contract when the guarantee is valuable.
- **Financial Model:** Using Black-Scholes underestimates tail risks (Fat Tails). Adding jumps (Jump-Diffusion) is recommended for long horizons.

8. AVAILABILITY AND REFERENCES

Resource availability

The R codes for sensitivity analyses, simulation algorithms, and the full research paper are available upon request from the author.

Selected Bibliography:

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